

☒ Easy (10 Questions)

1.

Which one needs a host program to spread?

- A) Worm
- B) Virus
- C) Trojan
- D) Rootkit

Answer: B

Explanation: Viruses attach to host files; worms spread independently.

2.

Which virus infects the boot sector?

- A) File Infector
- B) Boot Sector Virus
- C) Macro Virus
- D) Polymorphic Virus

Answer: B

Explanation: Boot sector viruses infect the system's boot code.

3.

A worm can spread:

- A) Only with user action
- B) Automatically over networks
- C) Only on USB drives
- D) Only by email attachments

Answer: B

Explanation: Worms replicate and spread without user interaction.

4.

Which virus type changes its code to avoid detection?

- A) Stealth Virus
- B) Polymorphic Virus
- C) Macro Virus
- D) Multipartite Virus

Answer: B

Explanation: Polymorphic viruses mutate their code.

5.

Macro viruses infect:

- A) System boot sectors
- B) Documents with macros
- C) Network routers
- D) Antivirus software

Answer: B

Explanation: Macro viruses target macro-enabled documents.

6.

Antivirus signature detection works by:

- A) Monitoring program behavior
- B) Matching known virus patterns
- C) Encrypting files
- D) Running in a sandbox

Answer: B

Explanation: Signature-based scanning looks for known virus code.

7.

Rootkits help viruses by:

- A) Encrypting files
- B) Hiding virus presence from OS
- C) Spreading worms
- D) Creating backups

Answer: B

Explanation: Rootkits hide malware activities.

8.

Heuristic analysis detects viruses by:

- A) Checking file size
- B) Analyzing suspicious code behavior
- C) Matching signatures
- D) Encrypting files

Answer: B

Explanation: Heuristics analyze potential threats even if unknown.

9.

Which is NOT a virus evasion technique?

- A) Polymorphism
- B) Sandboxing
- C) Packing
- D) Code Obfuscation

Answer: B

Explanation: Sandboxing is a detection method, not evasion.

10.

Which virus resides in memory to infect files?

- A) Resident Virus
- B) Multipartite Virus
- C) Boot Sector Virus
- D) Macro Virus

Answer: A

Explanation: Resident viruses embed in RAM.

☒ Medium (20 Questions)

11.

What is the main difference between a virus and a worm?

- A) Virus spreads without user action; worm needs user
- B) Virus needs a host program; worm spreads independently
- C) Virus infects networks only; worm infects files only
- D) Virus cannot replicate; worm replicates

Answer: B

Explanation: Worms spread independently; viruses need host.

12.

Multipartite viruses:

- A) Infect files only
- B) Infect boot sector only
- C) Infect both files and boot sector
- D) Infect emails only

Answer: C

Explanation: Multipartite viruses attack multiple system parts.

13.

Encryption as an evasion technique:

- A) Makes virus faster
- B) Hides virus code until runtime
- C) Detects antivirus programs
- D) Deletes system files

Answer: B

Explanation: Encryption conceals code until executed.

14.

Stealth viruses:

- A) Slow down system speed
- B) Hide modifications from detection tools
- C) Spread over networks automatically
- D) Only infect macros

Answer: B

Explanation: They hide presence by intercepting system calls.

15.

Which virus type specifically targets document macros?

- A) Boot Sector Virus
- B) File Infector Virus
- C) Macro Virus
- D) Polymorphic Virus

Answer: C

Explanation: Macro viruses exploit macro scripts.

16.

Signature-based detection cannot detect:

- A) Known viruses
- B) Unknown or new viruses without signature
- C) Polymorphic viruses always
- D) Both B and C

Answer: D

Explanation: Signature detection fails with unknown or mutated viruses.

17.

Behavioral detection focuses on:

- A) Static code analysis
- B) Monitoring actions during program execution
- C) Signature matching
- D) File size only

Answer: B

Explanation: Detects suspicious runtime behaviors.

18.

Which method is most effective for zero-day virus detection?

- A) Signature scanning
- B) Heuristic analysis
- C) File deletion
- D) Firewall blocking

Answer: B

Explanation: Heuristics detect unknown malware by behavior.

19.

A worm causes:

- A) Network congestion and bandwidth consumption
- B) File corruption only
- C) System file verification errors
- D) Macro script errors

Answer: A

Explanation: Worms spread rapidly over networks.

20.

Which of these is NOT a virus detection technique?

- A) Sandboxing
- B) Heuristic Analysis
- C) Packet Sniffing
- D) Signature Scanning

Answer: C

Explanation: Packet sniffing monitors network traffic, not virus detection.

21.

Which evasion technique modifies malware each time it spreads?

- A) Polymorphism

- B) Rootkit
- C) Packing
- D) Code signing

Answer: A

Explanation: Polymorphic viruses mutate themselves.

22.

Which is TRUE about boot sector viruses?

- A) Infect executable files only
- B) Infect startup code, loading before OS
- C) Infect email attachments
- D) Infect macros only

Answer: B

Explanation: They load early, before OS startup.

23.

Antivirus heuristic engines:

- A) Depend on virus signature database
- B) Use rules and patterns to detect suspicious code
- C) Only scan emails
- D) Delete all unknown files

Answer: B

Explanation: Heuristics find suspicious patterns.

24.

A virus that rewrites its own code every time it infects is:

- A) Stealth Virus
- B) Metamorphic Virus
- C) Resident Virus
- D) Multipartite Virus

Answer: B

Explanation: Metamorphic viruses rewrite themselves completely.

25.

Packing helps viruses by:

- A) Compressing or encrypting code to evade signature detection
- B) Slowing down execution
- C) Removing antivirus
- D) Sending spam emails

Answer: A

Explanation: Packing hides the malware signature.

26.

Which virus type requires user to open an infected file?

- A) Worm
- B) Virus
- C) Rootkit
- D) None

Answer: B

Explanation: Virus needs user action to spread.

27.

Which is a typical symptom of a virus infection?

- A) Slower system performance
- B) Increased network traffic only
- C) Faster downloads
- D) System shutdown only

Answer: A

Explanation: Viruses often degrade system performance.

28.

What is the role of antivirus sandboxes?

- A) Run suspicious programs isolated to observe behavior
- B) Delete all files immediately
- C) Speed up virus scanning
- D) Encrypt files on disk

Answer: A

Explanation: Sandboxes observe program behavior safely.

29.

Which virus evasion technique involves hiding in OS kernel?

- A) Rootkits
- B) Packing
- C) Polymorphism
- D) Heuristics

Answer: A

Explanation: Rootkits embed in OS to hide malware.

30.

Which is NOT a characteristic of worms?

- A) Self-replicate without user interaction
- B) Attach to executable files
- C) Spread over networks
- D) Can cause network congestion

Answer: B

Explanation: Worms don't need to attach to files.

☒ Hard (10 Questions)

31.

Metamorphic viruses differ from polymorphic viruses by:

- A) Encrypting code each time

- B) Completely rewriting their code instead of just mutating
- C) Infecting only boot sectors
- D) Being slower to propagate

Answer: B

Explanation: Metamorphic viruses rewrite code entirely.

32.

Which detection method can identify packed viruses despite their obfuscation?

- A) Signature scanning only
- B) Heuristic and behavioral analysis
- C) File deletion
- D) Packet inspection

Answer: B

Explanation: Heuristics analyze behavior, bypassing packing.

33.

Which is the primary target of macro viruses?

- A) System boot sector
- B) Document files with macros
- C) Email clients
- D) Network routers

Answer: B

Explanation: Macro viruses exploit document macro languages.

34.

An antivirus uses cloud-based detection because:

- A) It increases local file scanning speed
- B) It accesses real-time updated virus signatures and analytics
- C) It runs faster on slow computers
- D) It deletes old files automatically

Answer: B

Explanation: Cloud-based systems get real-time updates.

35.

Which of the following best describes rootkit capabilities?

- A) Encrypt data before transmission
- B) Hide files, processes, and network connections from OS and security software
- C) Delete all system backups
- D) Scan for viruses automatically

Answer: B

Explanation: Rootkits stealthily hide malware.

36.

Packing and encryption are used together because:

- A) Packing compresses code, encryption hides content
- B) Both delete system files
- C) Both speed up virus execution
- D) Both uninstall antivirus

Answer: A

Explanation: Packing compresses, encryption obfuscates.

37.

Which virus detection method is most effective for polymorphic viruses?

- A) Signature-based scanning
- B) Behavioral and heuristic analysis
- C) File size checking
- D) Disk defragmentation

Answer: B

Explanation: Behavioral and heuristic methods detect mutated viruses.

38.

What makes boot sector viruses particularly dangerous?

- A) They load before OS, evading many antivirus programs at startup
- B) They can't be detected by any antivirus
- C) They only infect USB drives
- D) They slow down internet speed

Answer: A

Explanation: Boot sector viruses execute early in the boot process.

39.

Why is antivirus evasion a serious challenge for security?

- A) It allows malware to avoid detection and persist longer
- B) It speeds up virus scanning
- C) It improves antivirus performance
- D) It prevents user training

Answer: A

Explanation: Evasion allows malware to remain hidden.

40.

What is the main limitation of signature-based antivirus?

- A) Cannot detect unknown or new malware without signatures
- B) Slows down system performance drastically
- C) Causes data loss
- D) Cannot scan emails

Answer: A

Explanation: Signature-based detection relies on known virus signatures.